# Vroomcheck · data-quality gate

The 19 checks between bronze (raw) and silver (staging) · `vroomcheck/ingestion/quality_checks.py` · last run 22 Oct 2026

raw · bronze
4 NHTSA tables, as text
→
**quality gate**
**19 checks**
→
dbt staging · silver
cleaned, typed views
·
critical fail → pipeline stops
dbt never runs

The gate runs on the raw tables after loading and before dbt builds staging. It does not change any data. It decides whether bronze is fit to build silver from. A failing **critical** check stops the pipeline. A failing **warning** check is logged and the run continues. Every result is appended to `raw._dq_results` with a timestamp, so data health can be tracked over time.

| 19 | 12 | 7 | 19 / 19 |
|---|---|---|---|
| checks per run | critical (stop the run) | warnings (logged only) | passed on both runs (21 and 22 Oct 2026) |

## Checks 1–4 · Complaints (raw_complaints)

**1** **row_count** CRITICAL — **1,114,946 ✓**

Counts the rows in the table. If the download or load silently failed, the table would be empty and everything downstream would be built from nothing.

Pass: at least 1 row

**2** **CMPLID_null_pct** CRITICAL — **0% ✓**

Percentage of complaints with a blank complaint ID. Without an ID a row can't be uniquely identified or traced back to NHTSA.

Pass: ≤ 0.5%

**3** **LDATE_unparseable_pct** CRITICAL — **0% ✓**

Percentage of received dates (LDATE) that aren't valid YYYYMMDD dates. Detection counts complaints per month, so a bad date means a lost complaint.

Pass: ≤ 5%

**4** **reject_pct** WARNING — **0% ✓**

Percentage of complaint file lines the loader couldn't parse (usually the wrong number of columns). Those lines go to `_load_rejects` instead of the table.

Pass: ≤ 1%

## Checks 5–8 · Recalls (raw_recalls)

**5** **row_count** CRITICAL — **246,253 ✓**

The recalls table isn't empty. Recalls are the ground truth for the back-test, so an empty table would make the 34.8% hit rate meaningless.

Pass: at least 1 row

6 **RECORD_ID_null_pct** CRITICAL — 0% ✓

Percentage of recall rows with a blank record ID.

Pass: ≤ 0.5%

7 **RCDATE_unparseable_pct** CRITICAL — 0% ✓

Percentage of recall report dates that aren't valid. The back-test measures lead time ("flagged 9 months before the recall"), so it needs this date.

Pass: ≤ 5%

8 **reject_pct** WARNING — 0% ✓

Percentage of recall file lines that couldn't be parsed.

Pass: ≤ 1%

## Checks 9–12 · Investigations (raw_investigations)

9 **row_count** CRITICAL — 154,439 ✓

The investigations table isn't empty.

Pass: at least 1 row

10 **ACTION_NUMBER_null_pct** CRITICAL — 0% ✓

Percentage of rows with a blank action number, the investigation ID (e.g. PE24-012).

Pass: ≤ 0.5%

11 **ODATE_unparseable_pct** CRITICAL — 0.10% ✓

Percentage of investigation opened dates that aren't valid. Used for the "11.1 months ahead of NHTSA's own investigations" result. The only critical check above 0: a few very old records have missing dates.

Pass: ≤ 5%

12 **reject_pct** WARNING — 0% ✓

Percentage of investigation file lines that couldn't be parsed.

Pass: ≤ 1%

## Checks 13–16 · Technical service bulletins (raw_tsbs)

13 **row_count** CRITICAL — 5,309,250 ✓

The TSB table isn't empty.

Pass: at least 1 row

14 **NHTSA_ID_null_pct** CRITICAL — 0% ✓

Percentage of bulletins with a blank NHTSA ID.

Pass: ≤ 0.5%

15 **MFR_COMM_DATE_unparseable_pct** CRITICAL — 0% ✓

Percentage of manufacturer communication dates that aren't valid.

Pass: ≤ 5%

16 **reject_pct** WARNING 0% ✓

Percentage of TSB file lines that couldn't be parsed.

Pass: ≤ 1%

## Checks 17–19 · Dataset-specific

17 **complaints · narrative_missing_pct** WARNING 0.49% ✓

Percentage of complaints whose narrative (CDESCR) is empty or under 20 characters. The narrative is what gets embedded and clustered, and what the AI explains. A complaint with no text can still be counted, but it can't be understood.

Pass: ≤ 5%

18 **complaints · future_received_dates** CRITICAL 0 ✓

Number of complaints with a received date after tomorrow. A complaint can't arrive in the future, so any hit means a corrupted file or a date-parsing error, and it would create a fake spike in the latest month.

Pass: exactly 0

19 **recalls · campno_format_bad_pct** WARNING 0.68% ✓

Percentage of recall campaign numbers that don't match NHTSA's format: 2 digits, a letter (V vehicle, E equipment, I, C child seat, T tire), then 3 digits, e.g. `24V123`. Campaign numbers link complaints, recalls and investigations. Mostly old, oddly formatted records.

Pass: ≤ 1%

## Summary table

| # | Dataset | Check | Pass rule | Type | Result |
|---|---|---|---|---|---|
| 1 | complaints | row_count | ≥ 1 | critical | 1,114,946 ✓ |
| 2 | complaints | CMPLID_null_pct | ≤ 0.5% | critical | 0% ✓ |
| 3 | complaints | LDATE_unparseable_pct | ≤ 5% | critical | 0% ✓ |
| 4 | complaints | reject_pct | ≤ 1% | warning | 0% ✓ |
| 5 | recalls | row_count | ≥ 1 | critical | 246,253 ✓ |
| 6 | recalls | RECORD_ID_null_pct | ≤ 0.5% | critical | 0% ✓ |
| 7 | recalls | RCDATE_unparseable_pct | ≤ 5% | critical | 0% ✓ |
| 8 | recalls | reject_pct | ≤ 1% | warning | 0% ✓ |
| 9 | investigations | row_count | ≥ 1 | critical | 154,439 ✓ |
| 10 | investigations | ACTION_NUMBER_null_pct | ≤ 0.5% | critical | 0% ✓ |
| 11 | investigations | ODATE_unparseable_pct | ≤ 5% | critical | 0.10% ✓ |
| 12 | investigations | reject_pct | ≤ 1% | warning | 0% ✓ |
| 13 | tsbs | row_count | ≥ 1 | critical | 5,309,250 ✓ |
| 14 | tsbs | NHTSA_ID_null_pct | ≤ 0.5% | critical | 0% ✓ |
| 15 | tsbs | MFR_COMM_DATE_unparseable_pct | ≤ 5% | critical | 0% ✓ |
| 16 | tsbs | reject_pct | ≤ 1% | warning | 0% ✓ |
| 17 | complaints | narrative_missing_pct | ≤ 5% | warning | 0.49% ✓ |
| 18 | complaints | future_received_dates | = 0 | critical | 0 ✓ |
| 19 | recalls | campno_format_bad_pct | ≤ 1% | warning | 0.68% ✓ |

## Two things to say accurately in the presentation

- **The gate decides, dbt cleans.** The gate only decides whether bronze is fit to build silver from. The actual cleaning (trimming, typing dates, conforming names, capping harm counts) happens in the dbt staging views, and the 64 dbt tests check silver and gold afterwards.
- **It runs on DuckDB only.** The Snowflake load doesn't run this gate. Instead, Snowflake raw was verified to match DuckDB column by column with 0 rejects, so it is covered indirectly.